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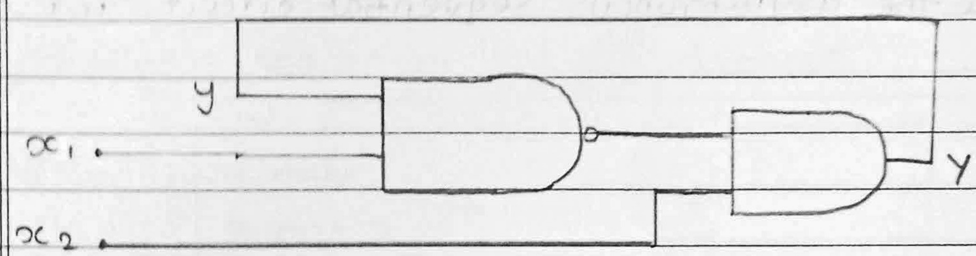
Course/Program:- BTech CSE SY
Subject:- DLM

DLM assignment 4.

Q.1) Analyze the circuit?

Solⁿ:-




① The first gate is a NAND gate with inputs y and $x_1$. The output of this gate is given by the Boolean expression:

$$\text{output}_1 = \overline{y \cdot x_1}$$

② The second gate is an AND gate. Its inputs are the output of the first gate and $x_2$. The output of this gate, which is the final output Y, is given by:

$$Y = \text{output}_1 \cdot x_2$$

③ Substitute the expression for output₁ into the equation for Y:

$$Y = \overline{y \cdot x_1} \cdot x_2$$

Using De Morgan's Theorem, which states that

$$\overline{A \cdot B} = \overline{A} + \overline{B}$$

We can rewrite expression as:

$$Y = (\bar{y} + \bar{x}_1) \cdot x_2$$

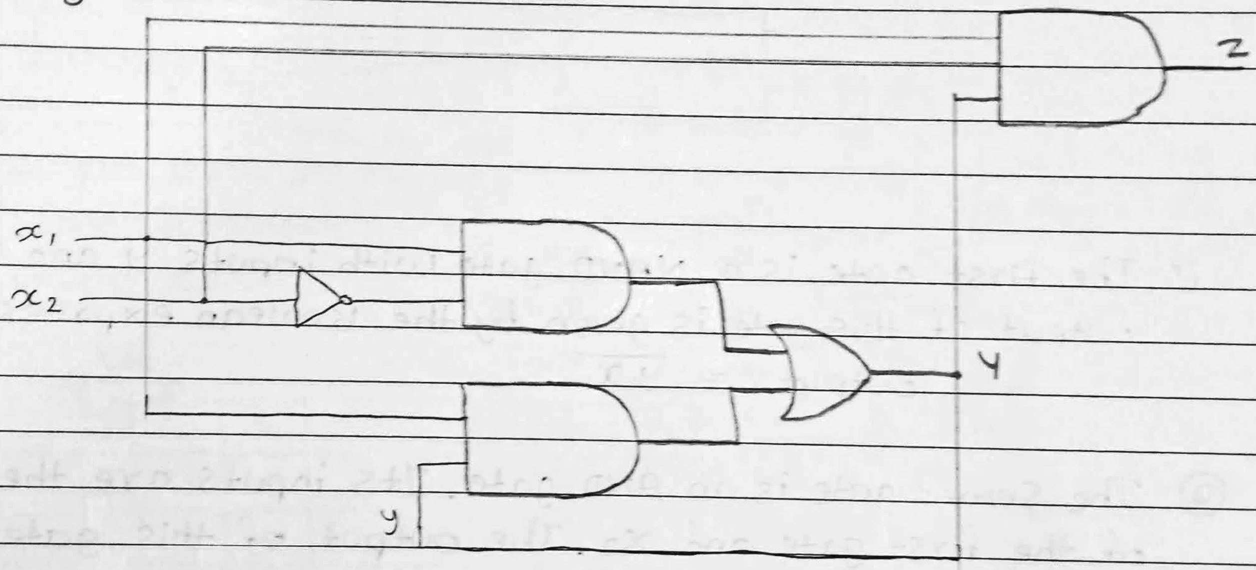
Distribute $x_2$

$$Y = \bar{y} . x_2 + \bar{x}_1 . x_2$$

→ The Boolean expression for the circuit is

$$Y = (\bar{y} + \bar{x}_1) \cdot x_2 \quad \text{or} \quad Y = \bar{y} . x_2 + \bar{x}_1 . x_2$$

Q.2) Analyze the asynchronous sequential circuit for given figure.




① NOT Gate
- Input : $x_2$
- output : $\bar{x}_2$
- This inverted signal is used in the first AND gate.

② First AND gate:
- Inputs: $x_1$ and $\bar{x}_2$
- outputs: $A_1 = x_1 . \overline{x_2}$

(3) Second AND gate:

- Inputs: $x_2$ and $y$
- Output: $x_2 \cdot y$

(4) OR gate:

- Inputs: $A_1$ and $A_2$
- outputs: $Y = A_1 + A_2 = x_1 \cdot \overline{x_2} + x_2 \cdot y$

This a feedback equation for output $y$, indicating that the circuit is sequential and asynchronous.

(5) Third AND gate:

- Inputs: $y$ and $x_2$
- outputs: $z = y \cdot x_2$

* Final Expressions

- output $Y$: $\boxed{Y = x_1 \cdot \overline{x_2} + x_2 \cdot y}$

This is a recursive equation, showing feedback from output $Y$ to its own input.

- output $\boxed{z = Y \cdot x_2}$

Q.3) Realize the circuit for which following transition table is ?

| $y_1y_2$ | $x$ = 0 | 1 |
|---|---|---|
| 00 | 00 | 01 |
| 01 | 11 | 01 |
| 11 | 11 | 10 |
| 10 | 00 | 10 |

→ Seperate the map of $Y_1$ and $Y_2$

for $Y_1$ :

| $y_1y_2$ \ $x$ | 0 | 1 |
|---|---|---|
| 00 | 0 | 1 |
| 01 | 1 | 0 |
| 11 | 1 | 1 |
| 10 | 0 | 1 |

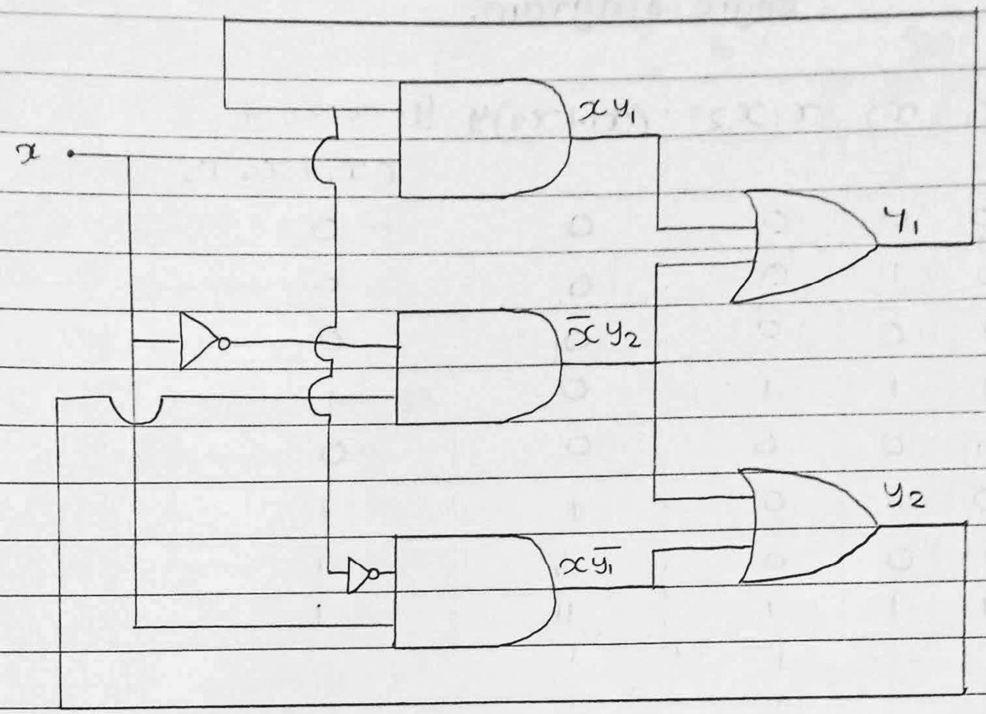
for $Y_2$:

| $y_1y_2$ \ $x$ | 0 | 1 |
|---|---|---|
| 00 | 0 | 1 |
| 01 | 1 | 1 |
| 11 | 1 | 0 |
| 10 | 0 | 0 |

Now write the boolean expression for $y_1$ and $y_2$

$$y_1 = xy_1 + \bar{x}y_2$$
$$y_2 = x\bar{y}_1 + \bar{x}y_2$$

Now we can draw the circuit from boolean expression.




Q.4) An asynchronous sequential circuit is described by the following excitation and output function.

$$Y = x_1x_2 + (x_1 + x_2)y$$

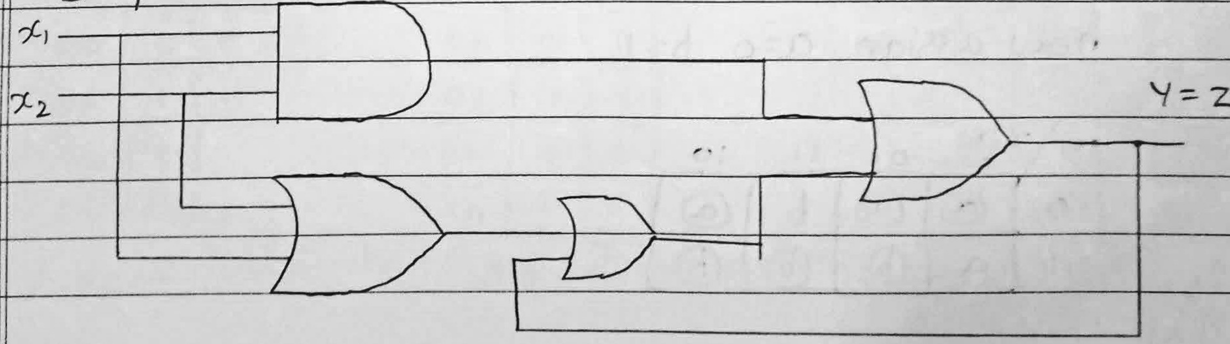
$$z = y$$

a) Draw the logic diagram of the circuit.
b) Derive the transition table, flow and output map.
c) Describe the behaviour of the circuit.

Sol$^n$:-

$$Y = x_1x_2 + (x_1 + x_2)y$$

$$z = Y$$



# Logic diagram.

| $y$ | $x_1$ | $x_2$ | $x_1x_2$ | $(x_1+x_2)y$ | $Y = x_1x_2 + (x_1+x_2)y$ |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 | 0 | 0 |
| 0 | 1 | 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 1 | 0 | 1 |
| 1 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 0 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 |

Transition Table:-

| $y \backslash x_1x_2$ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 0 |
| 1 | | 1 | 1 | 1 |

← (circle the stable state)

output map:

| $y \backslash x_1x_2$ | | | | |
|---|---|---|---|---|
| | 0 | 0 | - | 0 |
| | - | 1 | 1 | 1 |

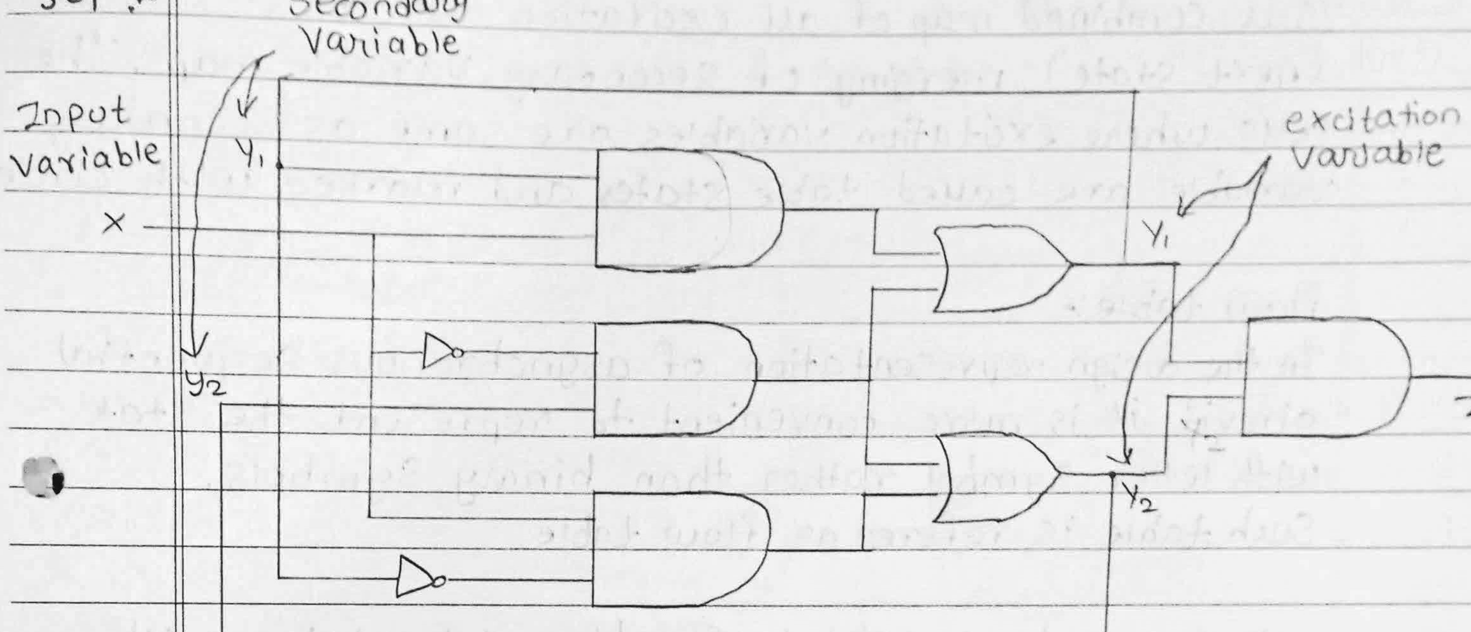
flow table:-

now assign $a=0$, $b=1$

| $y \backslash x_1x_2$ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | a | a | b | a |
| 1 | a | b | b | b |

Q.5) Analyze the asynchronous sequential circuit.

Sol$^n$:-




$y_1, y_2$ : Secondary variables

$Y_1, Y_2$ : Excitation variables

$\rightarrow Y_1 = x y_1 + \bar{x} y_2$

$Y_2 = x \bar{y}_1 + \bar{x} y_2$

$Z = Y_1 Y_2 = \bar{x} y_2$

Map: are similar to K-map with value of secondary variables (present state) labelling rows and value of input variables labelling columns and each cell displays value of excitation variable as per boolean function.

| $y_1y_2 \backslash x$ | 0 | 1 |
|---|---|---|
| 00 | 0 | 0 |
| 01 | 1 | 0 |
| 11 | 1 | 1 |
| 10 | 0 | 1 |

Map of $Y_1$

| $y_1y_2 \backslash x$ | 0 | 1 |
|---|---|---|
| 00 | 0 | 1 |
| 01 | 1 | 1 |
| 11 | 1 | 0 |
| 10 | 0 | 0 |

Map of $Y_2$

| $y_1y_2 \backslash x$ | 0 | 1 |
|---|---|---|
| 00 | 00 | 01 |
| 01 | 11 | 01 |
| 11 | 11 | 10 |
| 10 | 00 | 10 |

Transition table of $Y_1 Y_2$

Transition table:-
Is a combined map of all excitation variables $Y = Y_1 Y_2 \ldots$ (next state), merging of secondary variable maps. The cells where excitation variables are same as secondary variable are called table states and marked with circle.

flow table:-
In the design representation of asynchronous sequential circuit it is more convenient to represent the state with letter symbol rather than binary symbols. Such table is refered as flow table.

output may be included in flow/transition table with state seperated by a comma, giving complete description of the circuit.

| $y_1y_2 \backslash x$ | 0 | 1 |
|---|---|---|
| 00 | 00 | 01 |
| 01 | 11 | 01 |
| 11 | 11 | 10 |
| 10 | 00 | 10 |

Transition table of $Y_1 Y_2$

| $y_1y_2 \backslash x$ | 0 | 1 |
|---|---|---|
| 00 | 0 | 0 |
| 01 | 1 | 0 |
| 11 | 1 | 0 |
| 10 | 0 | 0 |

output map Z

| | $y_1y_2 \backslash x$ | 0 | 1 |
|---|---|---|---|
| a | ~~00~~ | a,0 | b,0 |
| b | ~~01~~ | c,1 | b,0 |
| c | ~~11~~ | c,1 | d,0 |
| d | ~~10~~ | a,0 | d,0 |

Flow table with output.